Engineering Document E17 — Storage, Knowledge & Data Lifecycle Architecture

Product Name: HQ

Engineering Package: Version 1.0

Purpose

This document defines how every piece of data within HQ is created, stored, classified, protected, retrieved, archived, and eventually deleted.

The objective is to establish a unified data architecture that supports AI intelligence, executive collaboration, enterprise security, and long-term scalability.

Data is one of HQ's most valuable assets and must be managed throughout its entire lifecycle.

Vision

Every piece of information inside HQ should have:

- A defined owner
- A storage location
- A security classification
- A lifecycle policy
- A retention period
- A backup strategy
- An audit trail

Nothing exists without governance.

Data Architecture

```
                    User Input
                        │
                        ▼
                 Data Classification
                        │
┌────────────┬──────────┴─────┬────────────┐
▼            ▼                ▼            ▼
PostgreSQL  Object Storage  Vector Store   Redis
│            │                │            │
└────────────┴──────────┬─────┴────────────┘
                        │
                        ▼
                  Knowledge Engine
                        │
                        ▼
                Mission Orchestrator
```

Storage Layers

HQ uses multiple specialized storage systems.

Layer 1 — Relational Storage

Technology:

- PostgreSQL
- Prisma ORM

Stores:

- Users
- Organizations
- Executives
- Missions
- Billing
- Authentication
- Permissions
- Audit Metadata

Structured business data belongs here.

Layer 2 — Object Storage

Stores:

- Images
- Videos
- PDFs
- Reports
- Presentations
- Exports
- Generated Assets
- Logos
- Attachments

Every object receives:

- UUID
- Owner
- Organization
- Security Level
- Version
- Hash
- Metadata

Layer 3 — Vector Database

Stores:

- Document embeddings
- Mission embeddings
- Knowledge embeddings
- Executive memories
- Organization memories

Used for:

- Semantic Search
- Retrieval-Augmented Generation (RAG)
- Context Retrieval
- Knowledge Discovery

Layer 4 — Redis

Stores:

- Sessions
- Queues
- Temporary execution state
- Cached analytics
- Prompt modules
- Active missions

Redis is never considered permanent storage.

Layer 5 — Audit Storage

Stores:

- Login history
- Executive decisions
- Permission changes
- Billing changes
- Organization updates
- Security events

Audit logs are immutable.

Data Classification

Every record belongs to one classification.

Public

Examples:

- Public templates
- Marketing assets
- Documentation

Internal

Examples:

- Analytics
- Non-sensitive configurations
- Internal reports

Confidential

Examples:

- Organization documents
- Executive discussions
- Mission outputs
- Business strategies

Restricted

Examples:

- Authentication
- Billing
- API Secrets
- Encryption Keys
- Compliance records

Restricted data receives maximum protection.

Data Ownership

Every record defines:

- Organization
- Owner

- Creator
- Last Modifier
- Created Date
- Updated Date

Ownership is never ambiguous.

File Lifecycle

```
Upload

↓

Validation

↓

Virus Scan

↓

Metadata Extraction

↓

Storage

↓

Indexing

↓

Usage

↓

Archive

↓

Deletion
```

Every uploaded file follows this lifecycle.

Knowledge Processing Pipeline

Uploaded documents pass through:

Upload

↓

Text Extraction

↓

OCR (if required)

↓

Chunking

↓

Embedding Generation

↓

Vector Storage

↓

Knowledge Library

↓

Executive Retrieval

Knowledge becomes searchable immediately.

Version Control

Supported for:

- Documents

- Brand Guidelines
- Policies
- SOPs
- Executive Templates

Users can:

- Compare versions
- Restore previous versions
- View history

No important document is permanently overwritten.

Data Retention Policies

Examples:

Working Memory

- Short-term

Mission Logs

- Long-term

Audit Logs

- Long-term

Temporary Uploads

- Automatic expiration

Retention policies are configurable per organization.

Backup Strategy

Automatic backups include:

- PostgreSQL
- Object metadata
- Vector indexes
- Configuration
- Audit logs

Backups are encrypted and validated.

Disaster Recovery

Recovery order:

Authentication

↓

Database

↓

Knowledge Library

↓

Object Storage

↓

Mission History

↓

Analytics

Core functionality is restored first.

Data Encryption

Encryption applies:

At Rest

- Database encryption
- Storage encryption

In Transit

- HTTPS
- TLS

Application Level

- Sensitive field encryption
- Secret management

Data Integrity

Every critical file includes:

- SHA-256 hash
- Upload timestamp
- Owner verification
- Version identifier

Integrity checks run automatically.

Search Architecture

Search supports:

- Full-text search
- Semantic search
- Hybrid search
- Executive search
- Mission search
- Organization search

Search indexes update automatically.

Archival Strategy

Older data transitions to archive storage.

Archive candidates:

- Completed missions
- Old reports
- Previous document versions
- Historical analytics

Archived data remains searchable if authorized.

Data Deletion

Deletion workflow:

Request

↓

Permission Check

↓

Dependency Validation

↓

Soft Delete

↓

Retention Window

↓

Permanent Deletion

This minimizes accidental loss.

Compliance Readiness

The architecture is designed to support:

- Data export
- Data deletion requests
- Audit reporting
- Organization isolation
- Retention management

This prepares HQ for future enterprise and regulatory requirements.

Storage Analytics

Monitor:

- Storage growth
- Vector index size
- Retrieval latency
- Cache hit ratio
- Backup success
- Archive utilization

These metrics support operational planning.

AI Integration

The storage architecture integrates with:

- Memory Engine
- Knowledge Library
- Prompt Composition Engine
- Executive Library

- Mission Orchestrator
- Analytics Platform

Every AI capability depends on reliable data management.

Future Evolution

Supports:

- Multi-region replication
- Cold storage tiers
- Enterprise Bring Your Own Storage (BYOS)
- External knowledge connectors
- Distributed vector databases
- Automated lifecycle optimization

No redesign is required for future expansion.

Success Criteria

The Storage, Knowledge & Data Lifecycle Architecture succeeds when:

- Every piece of data has a defined storage location.
- Knowledge is immediately retrievable by authorized executives.
- Data remains secure, versioned, and auditable.
- Backup and recovery processes are reliable.
- The platform scales without compromising governance or performance.

Conclusion

The HQ Storage, Knowledge & Data Lifecycle Architecture provides a comprehensive framework for managing data from creation to deletion.

By combining relational storage, object storage, vector databases, caching, immutable audit logs, lifecycle policies, and strong governance, HQ establishes a trusted data foundation that enables intelligent AI collaboration, enterprise-grade security, and long-term organizational learning.